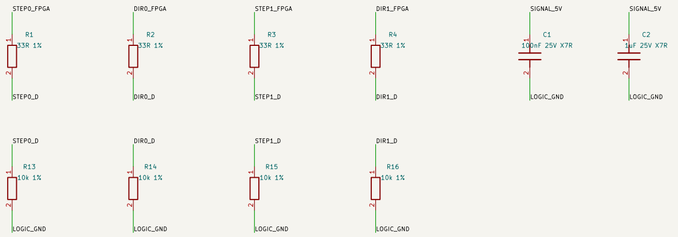
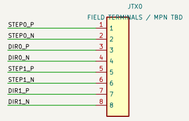
BASELINE SIX-JOINT DIFFERENTIAL TRANSMITTER — SCHEMATIC BLOCK r0.2

12 STEP/DIR signals / 3 AM26C31IPWR banks. No PCB, line protection, external permission latch or machine qualification.

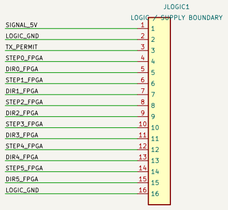
BANK TX0 / U1: JOINTS 0 AND 1



Place C1 close to U1 pins 16/8.
JTX0 matches ledger positions 1-8: reference is separate.

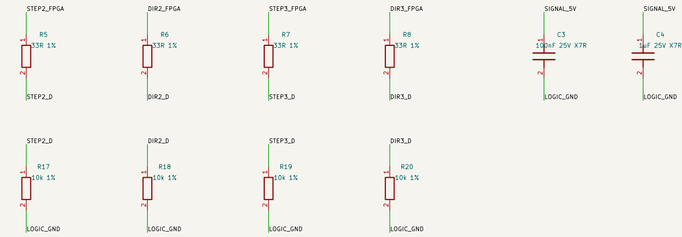


TX_PERMIT must be a hardware-qualified 3.3V permission signal. /G is tied high: G has three 10k pulldowns.
Absent permission means HIGH-Z, not a guaranteed low at the remote receiver. Independent drive inhibit and receiver bias are required.

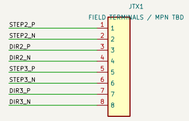


JLOGIC1 is an internal circuit boundary,
not an assigned FPGA package or connector.
3.3V logic; no 5V pull-ups on FPGA nets.
Do not drive inputs while drivers are unpowered.
SIGNAL_5V is externally supplied at 4.5-5.5V.
No current, thermal or partial-power proof.

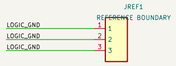
BANK TX1 / U2: JOINTS 2 AND 3

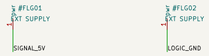


Place C3 close to U2 pins 16/8.
JTX1 matches ledger positions 1-8: reference is separate.

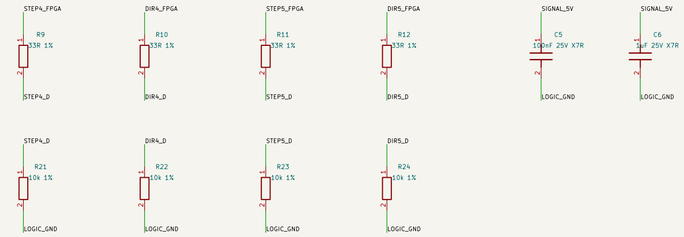


JREF1.1/2/3 map to ledger JPOWER.5/6/7:
TX0/1/2_REFERENCE, each bonded to LOGIC_GND.
JREF1 is only a review boundary, not a new final connector.
Line pairs are not galvanically isolated; qualify common mode.

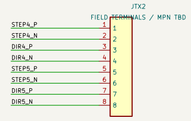




BANK TX2 / U3: JOINTS 4 AND 5



Place C5 close to U3 pins 16/8.
JTX2 matches ledger positions 1-8: reference is separate.



TI SLLS103P: Fig 4-1, Table 7-1, Sections 5.3/8.3.
33R damping and 10k defaults remain provisional.
Three permission pulldowns load the source in parallel.
All capacitor/resistor MPNs and connector footprints TBD.
No fabrication files or manufacturing release.
Original two-joint tx_bank evidence is retained separately.

Partial schematic only; not released for manufacture

Sheet: /		
File: tx_six.kicad_sch		
Title: Baseline six-joint differential transmitter		
Size: A2	Date: 2026-09-28	**Rev: r0.2**
KiCad E.D.A. 10.0.4		Id: 1/1